

ATHER RIZTA



Ather Rizta S scooter Performance Analysis

Comprehensive data analysis of how Tire Pressure and Ride modes affects Vehicle performance.

Presented by:

Naveen Lara Lawrence | [✉ naveenlara7411@gmail.com](mailto:naveenlara7411@gmail.com)

Made with **GAMMA**

Project Summary

To analyze how tyre pressure, battery metrics, and ride modes affect electric scooter performance, specifically focusing on efficiency, speed, and range.

25

Days recorded telemetry data of the electric vehicle is simulated,each the rider commutes 2 days

18000

Records ,recorded at 5 sec interval

AIS-137

aligns with CAFE norms and AIS137 standards

Problem Statement

The study aims to:

- Identify correlations among tyre pressure, motor load, discharge current, and SOC drop.
- Evaluate performance consistency under real-world noise, degradation, and sensor drift.
- Align with India's **CAFE (Corporate Average Fuel Efficiency)** and **AIS-137 Part 3** standards for CO₂ reduction compliance.

Out of Scope

Mostly tyre pressure is not considered while evaluating the performance of the vehicle,but it causes Heavy drag that increases the rolling resistance , which leads to increased motor load,that affects the entire engineering chain.

Vehicle Specifications:

- **Model:** Ather Rizta S
- **Make:** Ather Energy
- **Drive:** Direct Drive
- **Motor Type:** PMSM (Permanent Magnet Synchronous Motor)
- **Battery Capacity:** 2.9 kWh
- **Battery Type:** NMC (Lithium Nickel Manganese Cobalt Oxide)
- **Top Speed:** 80 km/hr
- **Modes:** ZIP / SMART ECO
- **Maximum Torque:** 22 Nm
- **Max Power:** 4300 W
- **Range (IDC):** 123 km
- **Charging Time (0–80%):** 4 hr 30 min
- **Battery Voltage:** 51.1 V
- **Kerb Weight:** 108 kg
- **Wheelbase:** 1295 mm
- **Tyres:** 90/90-12 (Front & Rear, Tubeless)
- **Brakes:** Disc with CBS + Regenerative Braking
- **Suspension:** Telescopic Fork (Front) / Monoshock (Rear)

Project Roadmap

Tools & Libraries used:

- Python (Pandas, NumPy)
- Used Mysql to Create StarSchema
- Apache Superset for interactive Vizualisation.
- Jupyter Notebook

1 Data Collection and Cleaning

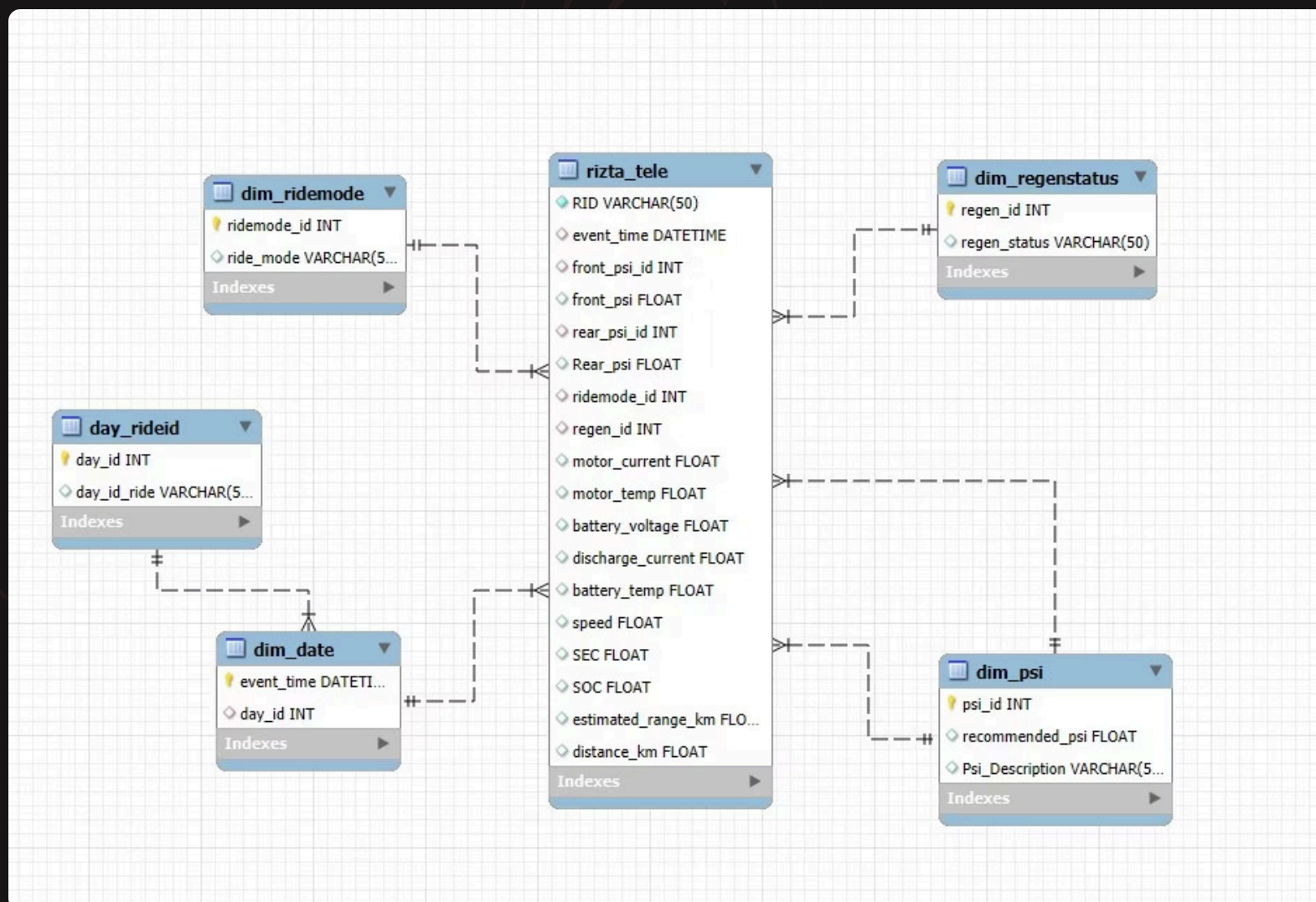
Simulated the Data using python with engineering logic, cleaned the data using pandas ,visualised and checked the outliers and capped them as per the skewness requirements.

2 Data Modelling:

Designed a star schema in my sql, used foreign key and primary key constraints, with indexes and views and created a database to connect with Apache superset

3 Dashboard:

Created a comprehensive dashboard that extracts insights about the telemetry data, created visualisations that tells the story about the performance degradation





Audience

Who

Engineering R&D team, Manager(R&D) and cross functional teams

Key Insight

Tyre pressure and tyre condition, battery performance, motor performance, scooter performance

Basis for key insight

Range ,acceleration, battery SOC.

Engineering Chain:

- **Rear_PSI ↓** : - Decrease in Rear PSI Increases Rolling Resistance
 - → **Motor Current ↑** - Increase in Load draws High Motor Current
 - → **Discharge Current ↑** - High Discharge of current from the battery
 - → **State of Charge (SOC) ↓ faster** - The Charge is drained faster
 - → **Battery Temperature ↑** - High Battery Temp leads to Battery Failure.
 - → **Battery Voltage ↓** - The High Temperature affects the battery voltage.
 - → **Specific Energy Consumed Per KM (Wh/km) ↓** - Which Increases the Energy Required for Km similar to milage.

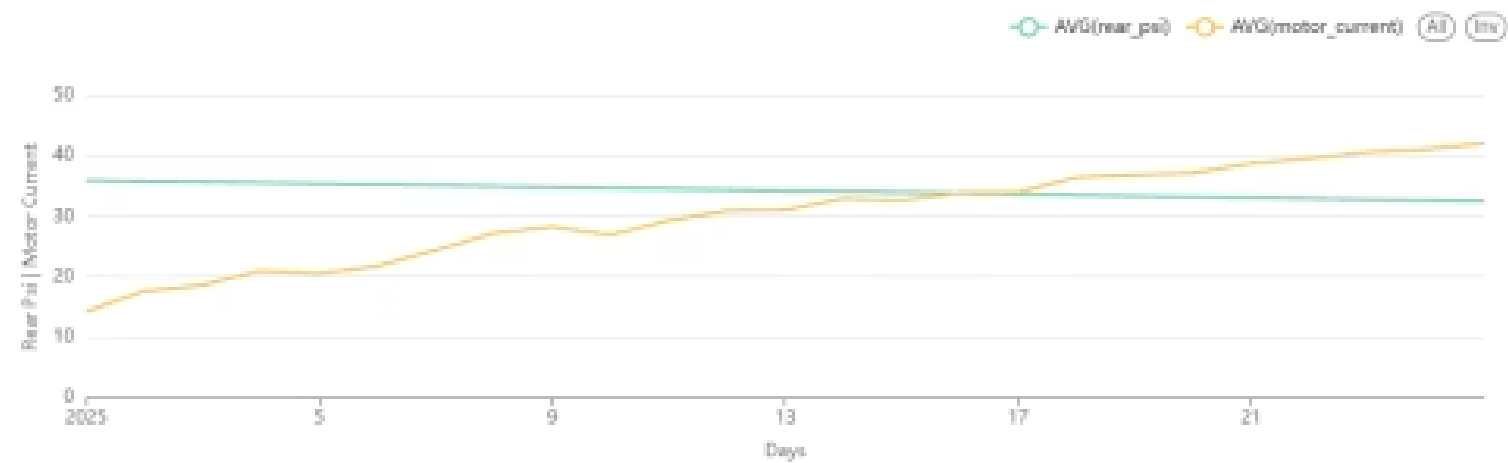
Low rear tire pressure increases rolling resistance, leading to higher current demand from the motor. This accelerates battery discharge, raises operating temperature, and ultimately reduces vehicle range and efficiency — mirroring real-world mileage drops observed in under-inflated tires.



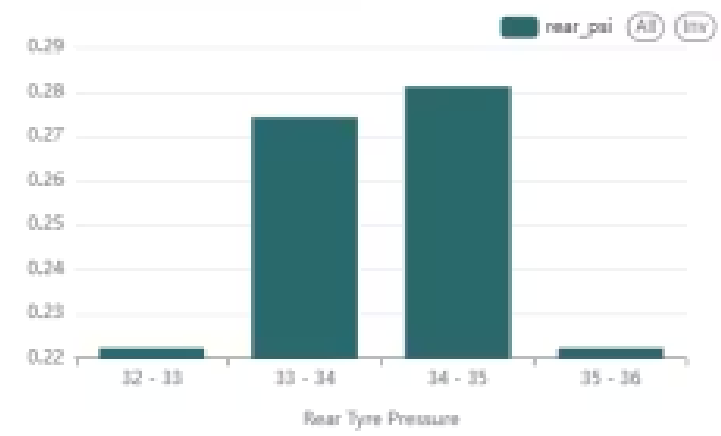
Correlation and Distribution Analysis

- The KPI cards highlight key exploratory metrics such as average rear tire pressure, motor current, SOC, and SEC.
- These indicators reveal that tire pressure is not just a simple maintenance parameter – it's a critical factor influencing the vehicle's overall energy efficiency.
- By connecting mechanical health with electrical performance, the KPI layer sets the stage for deeper analysis in the subsequent tabs.

Rear Psi vs Motor Current

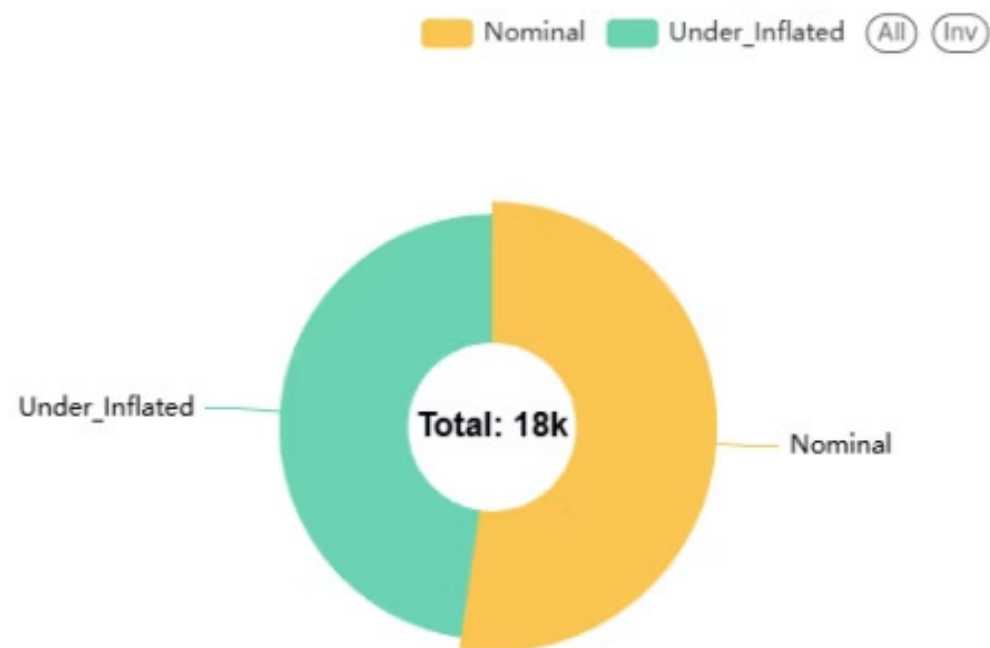


Rear Tyre Pressure Distribution



- Navigating to the Line chart, revealing the rear tire pressure decreases over time that causes the increase in motor current.
- The bar chart explains about the most rides cluster around the specific pressure ranges.

PSI CATEGORY

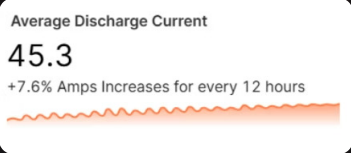


- The Pressure ranges are categorized into **NOMINAL and UNDER INFLATED**, which leads to Specific energy consumption over tire pressure.
- The Box Plots explains the difference between the Quartile range differences of Specific Energy consumption between Nominal and under inflated tire pressures.

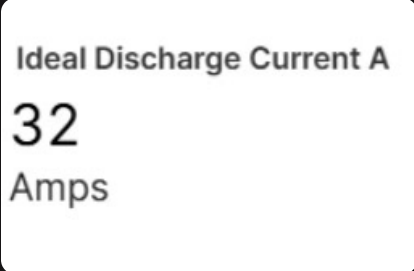
Energy Consumption vs PSI category



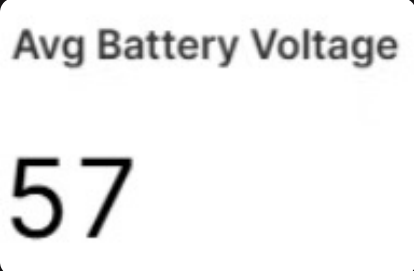
Performance Metrics over Time



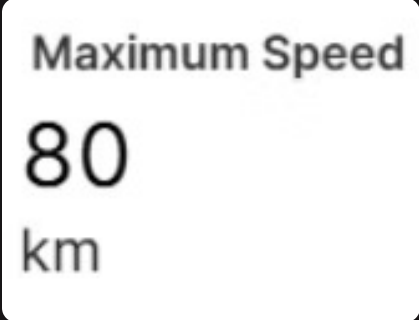
The average current consumed for a km (Mileage) increased for every 12 hrs in the dataset



But the ideal discharge current is only 32 amps should be consumed per km



The average battery voltage consumed is 57 V

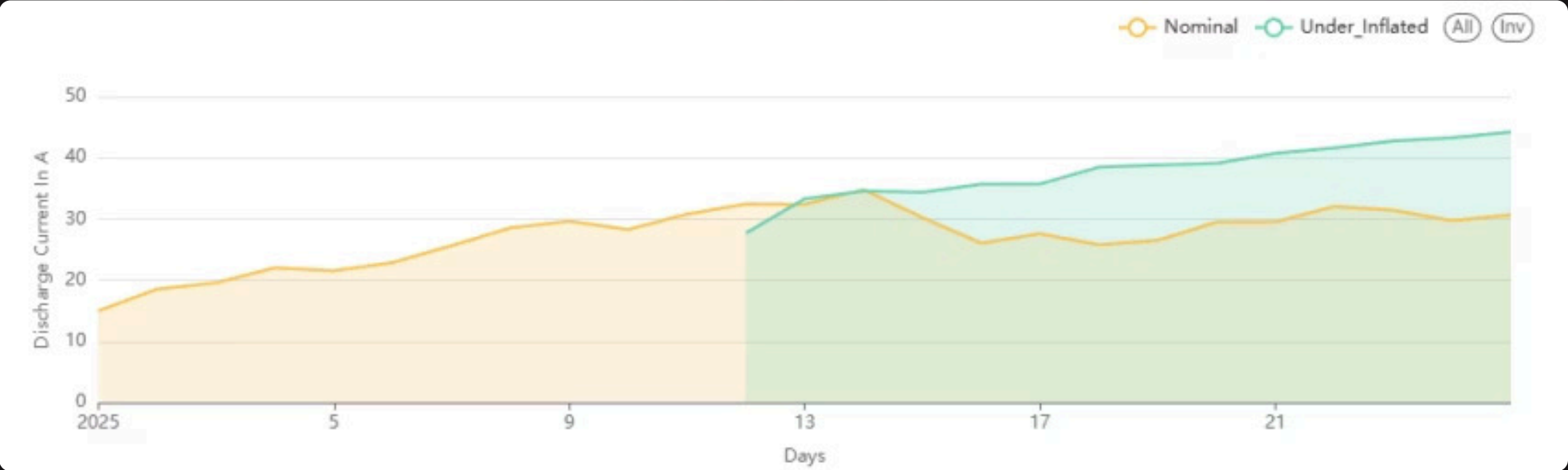
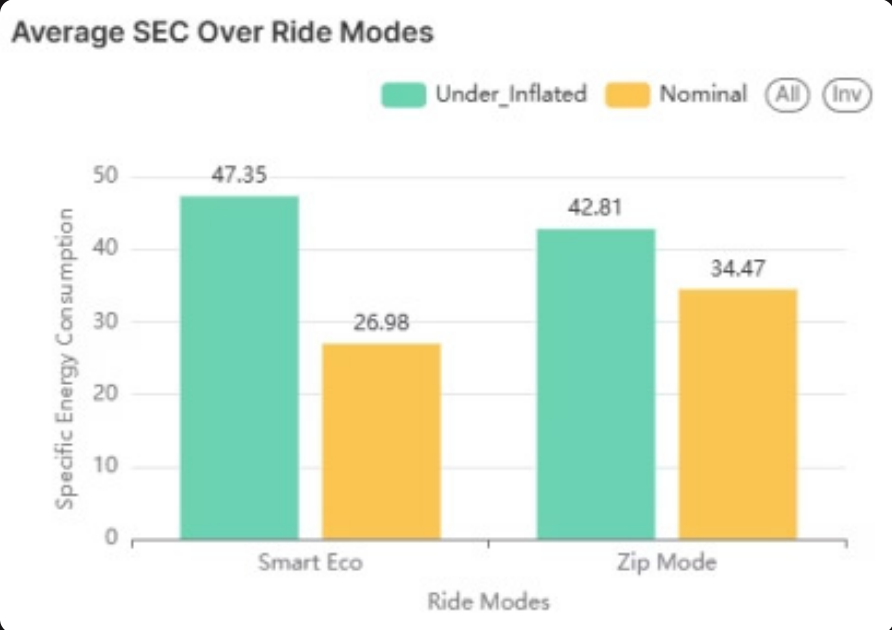


The Maximum Speed the vehicale can go is upto 80 km.

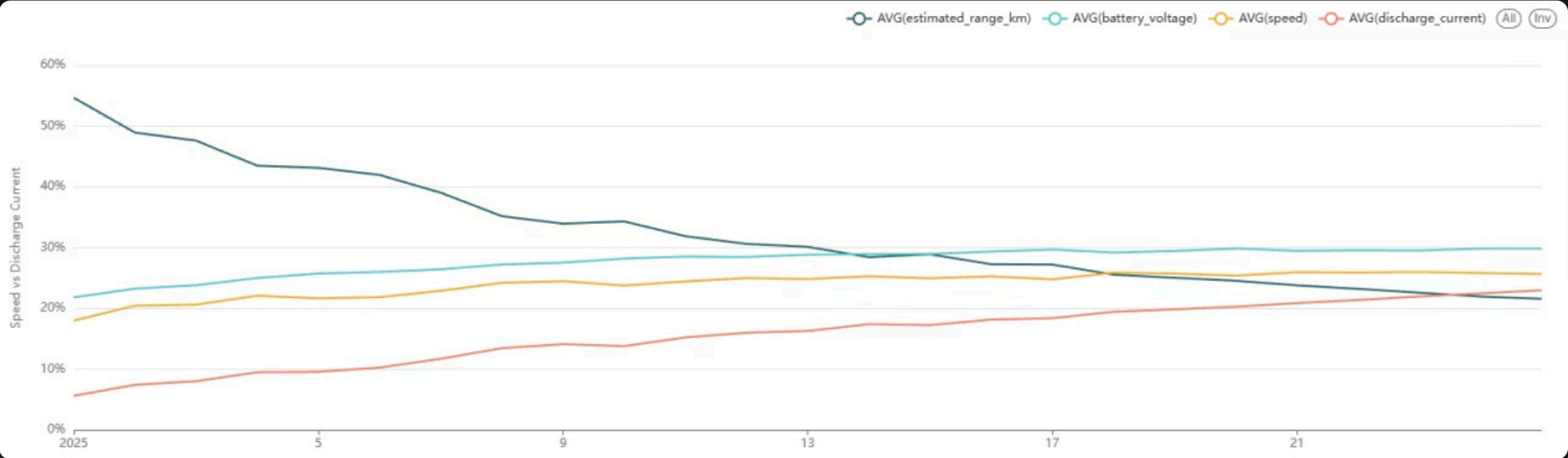


The Gauge chart explains about the speed difference in Zip and ECO mode, The ZIP is Faster compared to Smart ECO mode.

The Yellow bar says What the ideal range should have be, but the opposite part is explained in the under inflated category for both the modes.

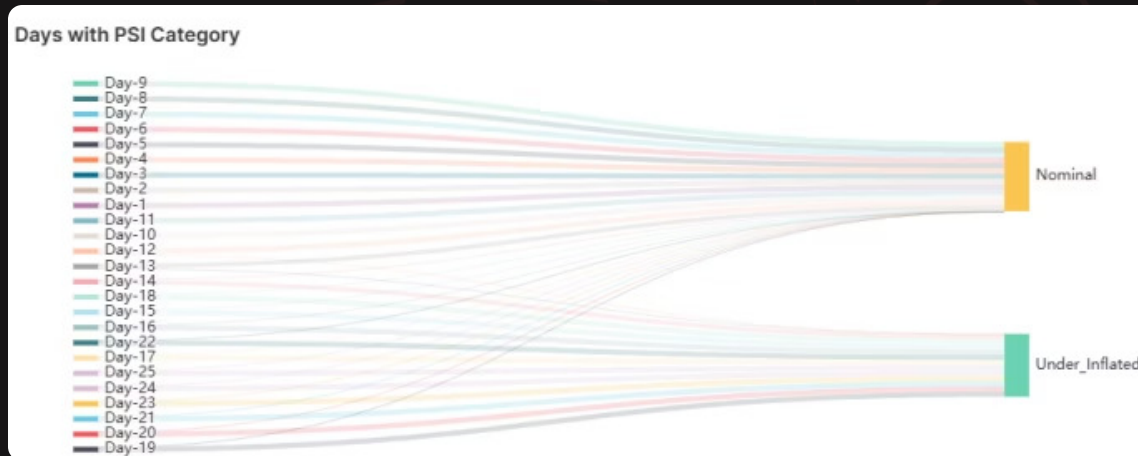
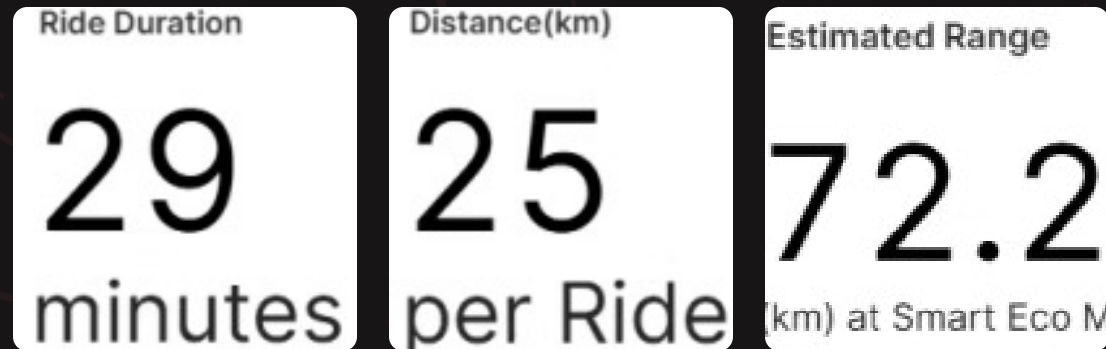


The Line chart clearly shows the Start of the under inflated tire trend before 13th day, which is suddenly inflated but the continuous leakage increases the current draw even more, spikes up to 44A.



The trend of the Range (green line)is reduced over the time due to increase in battery voltage,speed,discharge current,which was drastically affected due to under inflation affected in the previous line chart before day 13.

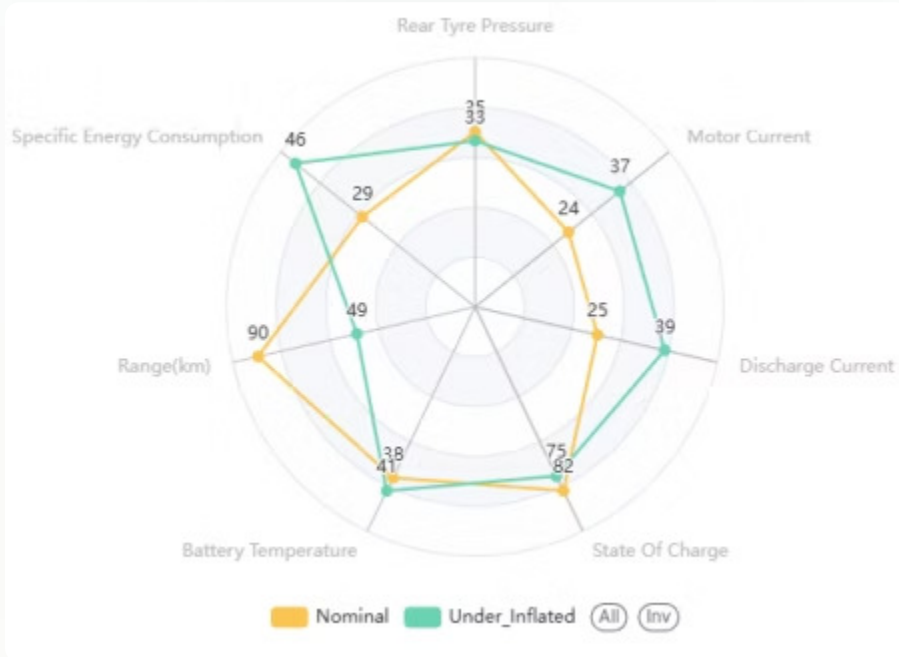
Categorical Analysis



The Sankey Chart explains aggregated distribution of tire, pressure categories with days as Source and Categories as Target, in which we can see the deflation starts before day 13.

Conclusion

- The radar chart illustrates how an often-overlooked parameter — rear tire pressure — influences the vehicle's core performance metrics across different ride modes.
- It emphasizes that optimizing tire pressure can directly enhance efficiency and range.
- To address performance drops and range anxiety, maintaining proper rear tire pressure should be treated as a key operational parameter — reminding us that **sometimes major performance improvements stem from the smallest adjustments.**



THANK YOU